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Problem Set 5: Comprehensive Solutions



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This problem set covers Chapters 21, 22, and 23 of Peng Ding's *A First Course in Causal Inference*, focusing on the Instrumental Variable (IV) method from both experimental and econometric perspectives.

1.1 Background: The Instrumental Variable Framework

Consider units $i = 1, \dots, n$. Let Z_i be the treatment assignment (instrument), D_i be the treatment received, and Y_i be the outcome.

Definition 1 (Latent Compliance Status) *Stratify the population into four latent groups based on $U_i = \{D_i(1), D_i(0)\}$:*

- *Always-taker (a):* $D_i(1) = 1, D_i(0) = 1$.
- *Complier (c):* $D_i(1) = 1, D_i(0) = 0$.
- *Defier (d):* $D_i(1) = 0, D_i(0) = 1$.
- *Never-taker (n):* $D_i(1) = 0, D_i(0) = 0$.

Assumption 1.1 (randomization) $Z \perp \{D(1), D(0), Y(1), Y(0)\}$.

Assumption 1.2 (monotonicity) $\Pr(U = d) = 0$, i.e., $D_i(1) \geq D_i(0)$ for all i .

Assumption 1.3 (exclusion restriction) $Y_i(1) = Y_i(0)$ for always-takers ($U = a$) and never-takers ($U = n$).

Theorem 1.1 (Theorem 21.1, Ding, 2025) *Under Assumptions ??-??, the Complier Average Causal Effect (CACE) is identified as:*

$$\tau_c = E\{Y(1) - Y(0) \mid U = c\} = \frac{E(Y \mid Z = 1) - E(Y \mid Z = 0)}{E(D \mid Z = 1) - E(D \mid Z = 0)}.$$

1.2 Exercises from Chapter 21: Compliance, IV, and LATE

21.1 Variance of the Wald estimator

Show that

$$\text{var}(\hat{\tau}_c) = \infty.$$

Solution to Problem 21.1: The Wald estimator is defined as $\hat{\tau}_c = \hat{\tau}_Y / \hat{\tau}_D$, where $\hat{\tau}_D = \bar{D}_1 - \bar{D}_0$ is the denominator. In any finite sample, there is a strictly positive probability that $\hat{\tau}_D = 0$ (for example, if all sampled units happen to be never-takers or always-takers such that the proportions are equal). When $\hat{\tau}_D = 0$, the ratio $\hat{\tau}_c$ is undefined. More formally, the variance involves the second moment $E[\hat{\tau}_c^2]$. Since $\hat{\tau}_D$ is a discrete random variable with $\text{pr}(\hat{\tau}_D = 0) > 0$, the expectation of the squared ratio involves a term with a zero denominator, causing the sum to diverge. Thus, $\text{var}(\hat{\tau}_c) = \infty$. This result underscores that the standard Wald estimator does not have finite moments in finite samples, even if the instrument is not "weak" in the population. $\square$

21.6 Binary IV and ordinal treatment received

Angrist and Imbens [1995] discussed a more general setting with a binary IV Z , an ordinal treatment received $D \in \{0, 1, \dots, J\}$, and outcome Y . The ordinal treatment received has potential outcomes $D(1)$ and $D(0)$, and the outcome has potential outcomes $Y(z, d)$. Under the following assumptions:

1. Randomization: $Z \perp \{D(z), Y(z, d) : z = 0, 1; d = 0, 1, \dots, J\}$;
2. Monotonicity: $D(1) \geq D(0)$;
3. Exclusion Restriction: $Y(z, d) = Y(d)$.

Prove that

$$\frac{E(Y | Z = 1) - E(Y | Z = 0)}{E(D | Z = 1) - E(D | Z = 0)} = \sum_{j=1}^J w_j E\{Y(j) - Y(j-1) | D(1) \geq j > D(0)\}$$

$$\text{where } w_j = \frac{\text{pr}\{D(1) \geq j > D(0)\}}{\sum_{j'=1}^J \text{pr}\{D(1) \geq j' > D(0)\}}.$$

Solution to Problem 21.6: We use the representation $D(z) = \sum_{j=1}^J \mathbb{I}(D(z) \geq j)$ and $Y(D(z)) = Y(0) + \sum_{j=1}^J [Y(j) - Y(j-1)] \mathbb{I}(D(z) \geq j)$.

The denominator of the Wald ratio is the ITT effect on D :

$$\begin{aligned} E[D(1) - D(0)] &= E \left[\sum_{j=1}^J (\mathbb{I}(D(1) \geq j) - \mathbb{I}(D(0) \geq j)) \right] \\ &= \sum_{j=1}^J \text{pr}(D(1) \geq j > D(0)), \end{aligned}$$

where the last equality follows from monotonicity $D(1) \geq D(0)$, which implies $\mathbb{I}(D(1) \geq j) - \mathbb{I}(D(0) \geq j)$ is a binary indicator for the event $D(1) \geq j > D(0)$. The numerator is the ITT effect on Y :

$$\begin{aligned} E[Y(D(1)) - Y(D(0))] &= E \left[\sum_{j=1}^J [Y(j) - Y(j-1)] (\mathbb{I}(D(1) \geq j) - \mathbb{I}(D(0) \geq j)) \right] \\ &= \sum_{j=1}^J E\{Y(j) - Y(j-1) \mid D(1) \geq j > D(0)\} \text{pr}(D(1) \geq j > D(0)). \end{aligned}$$

Dividing the numerator by the denominator, we obtain the weighted average:

$$\frac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{E[D \mid Z = 1] - E[D \mid Z = 0]} = \frac{\sum_{j=1}^J E\{Y(j) - Y(j-1) \mid D(1) \geq j > D(0)\} \text{pr}(D(1) \geq j > D(0))}{\sum_{j'=1}^J \text{pr}(D(1) \geq j' > D(0))}.$$

The weights w_j are exactly as defined in the problem. $\square$

21.7 Data analysis: a flu shot encouragement design

The dataset in `fludata.txt` is from a randomized encouragement design of McDonald et al. [1992]. It contains variables: `assign` (Z), `receive` (D), `outcome` (Y), and covariates (`age`, `sex`, `race`, `copd`, `dm`, `heartd`, `renal`, `liverd`). Analyze the data with and without adjusting for the covariates.

Solution to Problem 21.7: The results of the analysis are summarized below. Standard errors were calculated using the delta method and bootstrap (not shown here for brevity).

- **Unadjusted Analysis (Wald Estimator):**

- $\hat{\tau}_Y$ (ITT on Outcome): -0.0147
- $\hat{\tau}_D$ (ITT on Treatment): 0.1184
- $\hat{\tau}_c$ (CACE): **-0.1246**

- **Adjusted Analysis (Lin Adjustment):**

- $\hat{\tau}_{Y,L}$: -0.0149
- $\hat{\tau}_{D,L}$: 0.1187

$$- \hat{\tau}_{c,L}: -0.1252$$

Interpretation: Encouraging patients to take a flu shot reduces the probability of flu-related hospitalization by approximately 12.5% among those who comply with the encouragement. Covariate adjustment slightly moves the estimate and generally improves precision. $\square$

1.3 Exercises from Chapter 22: IV Inequalities

Theorem 1.2 (Theorem 22.1, Ding, 2025) *Under Assumptions ??-??, the potential outcome means for compliers are identified as:*

$$\mu_{1c} = \frac{E(DY \mid Z = 1) - E(DY \mid Z = 0)}{\pi_c}, \quad \mu_{0c} = \frac{E\{(1 - D)Y \mid Z = 0\} - E\{(1 - D)Y \mid Z = 1\}}{\pi_c}$$

where $\pi_c = E(D \mid Z = 1) - E(D \mid Z = 0)$.

22.2 Risk ratio for compliers

With a binary outcome, define the risk ratio for compliers as $rr_c = \frac{\text{pr}\{Y(1)=1|U=c\}}{\text{pr}\{Y(0)=1|U=c\}}$. Show:

$$rr_c = \frac{E(DY \mid Z = 1) - E(DY \mid Z = 0)}{E\{(D - 1)Y \mid Z = 1\} - E\{(D - 1)Y \mid Z = 0\}}.$$

Solution to Problem 22.2: From Theorem ??, the numerator of rr_c is $\mu_{1c} = \frac{E(DY|Z=1)-E(DY|Z=0)}{\pi_c}$. The denominator is $\mu_{0c} = \frac{E\{(1-D)Y|Z=0\}-E\{(1-D)Y|Z=1\}}{\pi_c}$.

Note that $(D - 1)Y = DY - Y$. Thus,

$$\begin{aligned} E\{(D - 1)Y \mid Z = 1\} - E\{(D - 1)Y \mid Z = 0\} &= [E(DY \mid Z = 1) - E(Y \mid Z = 1)] - [E(DY \mid Z = 0) - E(Y \mid Z = 0)] \\ &= [E(Y \mid Z = 0) - E(DY \mid Z = 0)] - [E(Y \mid Z = 1) - E(DY \mid Z = 1)] \\ &= E\{(1 - D)Y \mid Z = 0\} - E\{(1 - D)Y \mid Z = 1\}. \end{aligned}$$

This is exactly the numerator of μ_{0c} (scaled by π_c). Taking the ratio μ_{1c}/μ_{0c} cancels π_c and yields the result. $\square$

22.7 Bounds on the average causal effect on the whole population

For a bounded outcome $Y \in [\underline{y}, \bar{y}]$, let $\delta = E\{Y(d = 1) - Y(d = 0)\}$. Prove the sharp bounds:

$$\underline{\delta} = \delta' - \bar{y}\text{pr}(D = 1 \mid Z = 0) + \underline{y}\text{pr}(D = 0 \mid Z = 1)$$

$$\bar{\delta} = \delta' - \underline{y}\text{pr}(D = 1 \mid Z = 0) + \bar{y}\text{pr}(D = 0 \mid Z = 1)$$

where $\delta' = E(DY \mid Z = 1) - E((1 - D)Y \mid Z = 0)$.

Solution to Problem 22.7: Decompose $\delta = \pi_a(m_{1a} - m_{0a}) + \pi_n(m_{1n} - m_{0n}) + \pi_c(m_{1c} - m_{0c})$, where $m_{du} = E[Y(d) \mid U = u]$. Identified terms are $\pi_a, \pi_n, \pi_c, m_{1a}, m_{0n}, m_{1c}, m_{0c}$. Unidentified terms are m_{0a} and m_{1n} . Define $\delta' = \pi_a m_{1a} + \pi_c(m_{1c} - m_{0c}) - \pi_n m_{0n}$. Then $\delta = \delta' + \pi_n m_{1n} - \pi_a m_{0a}$. Using $m_{1n}, m_{0a} \in [\underline{y}, \bar{y}]$: The lower bound is achieved when $m_{1n} = \underline{y}$ and $m_{0a} = \bar{y}$: $\underline{\delta} = \delta' + \pi_n \underline{y} - \pi_a \bar{y} = \delta' + \text{pr}(D = 0 \mid Z = 1)\underline{y} - \text{pr}(D = 1 \mid Z = 0)\bar{y}$. The upper bound is achieved when $m_{1n} = \bar{y}$ and $m_{0a} = \underline{y}$: $\bar{\delta} = \delta' + \pi_n \bar{y} - \pi_a \underline{y} = \delta' + \text{pr}(D = 0 \mid Z = 1)\bar{y} - \text{pr}(D = 1 \mid Z = 0)\underline{y}$. These are the sharp bounds. $\square$

1.4 Exercises from Chapter 23: Econometric Perspective and TSLS

23.1 More algebra for TSLS

1. Show that $\hat{\Gamma}$ in the first-stage OLS of D on Z equals $\hat{\Gamma} = (\sum_{i=1}^n Z_i Z_i^T)^{-1} \sum_{i=1}^n Z_i D_i^T$.
2. Show that $\hat{\beta}_{\text{TSLS}} = \hat{\beta}_{\text{IV}}$ in the just-identified case.

Solution to Problem 23.1: 1. The first stage regresses D on Z using OLS. The objective is to minimize $\sum_{i=1}^n \|D_i - \Gamma^T Z_i\|^2$. The normal equation is $\sum_{i=1}^n Z_i (D_i - \hat{\Gamma}^T Z_i)^T = 0$. Transposing and solving for $\hat{\Gamma}$ gives $\hat{\Gamma} = (\sum Z_i Z_i^T)^{-1} \sum Z_i D_i^T$.

2. In the just-identified case, Z and D have the same dimension. $\hat{\Gamma}$ is a square and invertible matrix. $\hat{D}_i = \hat{\Gamma}^T Z_i$. Thus $\sum \hat{D}_i \hat{D}_i^T = \hat{\Gamma}^T (\sum Z_i Z_i^T) \hat{\Gamma}$. Substituting $\hat{\Gamma}^T = (\sum D_i Z_i^T) (\sum Z_i Z_i^T)^{-1}$: $\sum \hat{D}_i \hat{D}_i^T = (\sum D_i Z_i^T) (\sum Z_i Z_i^T)^{-1} (\sum Z_i Z_i^T) (\sum Z_i Z_i^T)^{-1} \sum Z_i D_i^T = (\sum D_i Z_i^T) (\sum Z_i Z_i^T)^{-1} (\sum Z_i D_i^T)$. Similarly, $\sum \hat{D}_i Y_i = \hat{\Gamma}^T \sum Z_i Y_i = (\sum D_i Z_i^T) (\sum Z_i Z_i^T)^{-1} \sum Z_i Y_i$. $\hat{\beta}_{\text{TSLS}} = [(\sum D_i Z_i^T) (\sum Z_i Z_i^T)^{-1} (\sum Z_i D_i^T)]^{-1} (\sum D_i Z_i^T) (\sum Z_i Z_i^T)^{-1} \sum Z_i Y_i$. Using the property $(ABC)^{-1} = C^{-1} B^{-1} A^{-1}$: $\hat{\beta}_{\text{TSLS}} = (\sum Z_i D_i^T)^{-1} (\sum Z_i Z_i^T) (\sum D_i Z_i^T)^{-1} (\sum D_i Z_i^T) (\sum Z_i Z_i^T)^{-1} \sum Z_i Y_i = (\sum Z_i D_i^T)^{-1} \sum Z_i Y_i = \hat{\beta}_{\text{IV}}$. $\square$

23.2 Equivalence between TSLS and ILS

Prove that, with a single endogenous treatment and a single IV, the indirect least squares (ILS) estimator $\hat{\beta}_{1,\text{ILS}} = \hat{\Gamma}_1 / \hat{\gamma}_1$ equals $\hat{\beta}_{1,\text{TSLS}}$.

Solution to Problem 23.2: Consider the reduced form $Y = \Gamma_0 + \Gamma_1 Z + \epsilon_1$ and $D = \gamma_0 + \gamma_1 Z + \epsilon_2$. The ILS estimator is $\hat{\beta}_{1,\text{ILS}} = \hat{\Gamma}_1 / \hat{\gamma}_1$. From OLS theory, $\hat{\gamma}_1 = \text{cov}(Z, D) / \text{var}(Z)$ and $\hat{\Gamma}_1 = \text{cov}(Z, Y) / \text{var}(Z)$. Thus $\hat{\beta}_{1,\text{ILS}} = \text{cov}(Z, Y) / \text{cov}(Z, D)$. The TSLS estimator regresses Y on the fitted values

$\hat{D} = \hat{\gamma}_0 + \hat{\gamma}_1 Z$. Let $\tilde{Z} = Z - \bar{Z}$. Then $\hat{D} - \bar{D} = \hat{\gamma}_1 \tilde{Z}$. $\hat{\beta}_{1,\text{TSLs}} = \frac{\text{cov}(\hat{D}, Y)}{\text{var}(\hat{D})} = \frac{\text{cov}(\hat{\gamma}_1 Z, Y)}{\text{var}(\hat{\gamma}_1 Z)} = \frac{\hat{\gamma}_1 \text{cov}(Z, Y)}{\hat{\gamma}_1^2 \text{var}(Z)} = \frac{\text{cov}(Z, Y)}{\hat{\gamma}_1 \text{var}(Z)} = \frac{\text{cov}(Z, Y)}{\frac{\text{cov}(Z, D)}{\text{var}(Z)} \text{var}(Z)} = \frac{\text{cov}(Z, Y)}{\text{cov}(Z, D)}$. Thus $\hat{\beta}_{1,\text{ILs}} = \hat{\beta}_{1,\text{TSLs}}$. $\square$

23.3 Control function in the linear IV model

Definition: The control function estimator $\hat{\beta}_{\text{CF}}$ is obtained by: (1) Regressing D on Z to get residuals $\check{D}_i$. (2) Regressing Y on D and $\check{D}$ using OLS. The coefficient of D is $\hat{\beta}_{\text{CF}}$. Show that $\hat{\beta}_{\text{CF}} = \hat{\beta}_{\text{TSLs}}$.

Solution to Problem 23.3: By the Frisch-Waugh-Lovell (FWL) theorem, the coefficient of D in a regression of Y on D and $\check{D}$ is the same as the coefficient in a regression of Y on the residuals of D after regressing it on $\check{D}$. We have $D = \hat{D} + \check{D}$, where $\hat{D}$ is the fitted value from Z and $\check{D}$ is the residual. Since $\hat{D}$ and $\check{D}$ are orthogonal by construction in OLS, the residual of D regressed on $\check{D}$ is simply $D - \check{D} = \hat{D}$. Therefore, $\hat{\beta}_{\text{CF}}$ is obtained by regressing Y on $\hat{D}$, which is exactly the definition of the second stage of TSLs. Thus, $\hat{\beta}_{\text{CF}} = \hat{\beta}_{\text{TSLs}}$. $\square$

1.5 Additional Problems

2. CACE as Ratio of Covariances

Show that $\text{CACE} = \frac{\text{cov}(Z, Y)}{\text{cov}(Z, D)}$.

Solution to Problem 2: For a binary instrument Z , let $p = \text{pr}(Z = 1)$. $\text{cov}(Z, Y) = E[ZY] - E[Z]E[Y] = pE[Y | Z = 1] - p(pE[Y | Z = 1] + (1-p)E[Y | Z = 0]) = p(1-p)[E[Y | Z = 1] - E[Y | Z = 0]]$. Similarly, $\text{cov}(Z, D) = p(1-p)[E[D | Z = 1] - E[D | Z = 0]]$. The ratio is $\frac{\text{cov}(Z, Y)}{\text{cov}(Z, D)} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]}$. By Theorem ??, this ratio identifies the CACE. $\square$

3. Weighting method for CACE in observational studies

Suppose randomization is replaced by conditional ignorability $Z \perp \{D(1), D(0), Y(1), Y(0)\} \mid X$, while monotonicity and exclusion restriction hold. Let $e(X) = \text{pr}(Z = 1 \mid X)$. Define $\kappa_{1i} = D_i \frac{Z_i - e(X_i)}{e(X_i)(1 - e(X_i))}$. Show that $E[\kappa_{1i}g(Y, X)] = \text{pr}(U = c)E[g(Y(1), X) \mid U = c]$.

Solution to Problem 3: Using the property of iterated expectations: $E[\kappa_{1i}g(Y, X)] = E_X[E[D_i \frac{Z_i - e(X_i)}{e(X_i)(1 - e(X_i))} g(Y, X) \mid X]]$. Under conditional ignorability: $E[D_i Z_i g(Y_i, X_i) \mid X_i] = E[Z_i \mid X_i]E[D_i(1)g(Y_i(1), X_i) \mid X_i] = e(X_i)E[D_i(1)g(Y_i(1), X_i) \mid X_i]$. $E[D_i e(X_i)g(Y_i, X_i) \mid X_i] =$

$e(X_i)[e(X_i)E[D_i(1)g(Y_i(1), X_i) | X_i] + (1 - e(X_i))E[D_i(0)g(Y_i(0), X_i) | X_i]]$. The numerator of the inner expectation is: $e(X)(1 - e(X))E[D(1)g(Y(1), X) | X] - e(X)(1 - e(X))E[D(0)g(Y(0), X) | X]$. Under exclusion restriction $Y(1) = Y(d = 1)$ and $Y(0) = Y(d = 0)$ for the relevant strata: The expression simplifies to $e(X)(1 - e(X))E[(D(1) - D(0))g(Y(1), X) | X]$. The factors $e(X)(1 - e(X))$ cancel out. Thus: $E[\kappa_{1i}g(Y, X)] = E[E[(D(1) - D(0))g(Y(1), X) | X]] = E[\mathbb{I}(U = c)g(Y(1), X)] = \text{pr}(U = c)E[g(Y(1), X) | U = c]$. A similar argument holds for κ_{0i} . $\square$

4. Analysis of Return to Schooling (Card, 1993)

Card [1993] used distance to college as an IV for education. Discuss the dichotomization approach vs. TSLS.

Solution to Problem 4:

- **Dichotomization:** One could define $D = 1$ if education $\geq c$ years. This estimates the return to reaching at least c years of education for those nudged by proximity. The choice of c (e.g., 12 for high school, 16 for college) changes the target latent group (compliers at different thresholds). Results may be sensitive if the "encouragement" effect varies across the education distribution.
- **TSLS:** Uses continuous variation in education. Under a linear structural model $Y = \beta_0 + \beta_1 D + \epsilon$, TSLS identifies β_1 . If the return is heterogeneous, TSLS identifies a weighted average of causal effects across the population (the "Average Derivative Effect").

In practice, TSLS is more common as it uses more information, but dichotomization provides a clearer interpretation of "who" the effect is being measured for (compliers at a specific level). $\square$



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